



# Long-term angiogenic and thromboinflammatory signatures in post-COVID-19 syndrome

Arlette Araceli Mora Zetina<sup>1,11</sup> · Erika Fabiola Ortega Sanchez<sup>1,11</sup> · Brenda Saraí Zuñiga Ascencio<sup>2,12</sup> · Armando Roberto Castorena Maldonado<sup>12</sup> · Víctor Hugo Ahumada Topete<sup>3,12</sup> · Luz María Torres Espíndola<sup>4</sup> · Arnoldo Aquino Gálvez<sup>5</sup> · Rosario Fernández Plata<sup>1,8</sup> · Anjarath Lorena Higuera Iglesias<sup>1,8</sup> · David Martínez Briseño<sup>1,8</sup> · Aranza Bernal Silva<sup>1</sup> · Misael Osmar García Martín<sup>12,13</sup> · José Eduardo Márquez García<sup>6</sup> · Angelica Moncada Morales<sup>6</sup> · Jaime Chávez Alderete<sup>7</sup> · Rafael Hernandez Zenteno<sup>8,12</sup> · Eduardo Becerril Vargas<sup>9,12</sup> · Daniela Josefina Cataneo Piña<sup>10,12</sup> · José Leopoldo Aguilar Faisal<sup>11,14</sup> · Jazmín García Machorro<sup>11,14</sup> · Pedro López Sánchez<sup>11,14</sup> · María Elena Hernandez Campos<sup>11,14</sup> · Manuel Castillejos López<sup>1,8,13</sup> 

Received: 8 May 2026 / Accepted: 10 July 2026  
© The Author(s) 2026

## Abstract

**Background** Post-COVID-19 syndrome is a major long-term sequela of severe SARS-CoV-2 infection, potentially involving persistent angiogenic and thromboinflammatory dysfunction, though long-term biomarker behavior remains unclear.

**Objective** To evaluate the evolution of angiogenic and thromboinflammatory biomarkers in post-COVID-19 syndrome patients.

**Methods** This ambispective cohort study was conducted at the National Institute of Respiratory Diseases in Mexico City. Thirty-two adults hospitalized for severe or critical COVID-19 in 2020 were followed for three and a half years. Paired plasma samples were collected during hospitalization and at long-term follow-up. Endothelial dysfunction markers included soluble P-selectin, vascular endothelial growth factor receptor 2, vascular endothelial growth factor D, and angiopoietin-1. Hemostasis was assessed by prothrombin time. Acute-phase proteins alpha-2-macroglobulin and haptoglobin were measured via immunoassay. Persistent symptoms were documented at follow-up.

**Results** At three and a half years, persistent symptoms were common: fatigue in sixty-five point 6%, dyspnea in sixty-two point 5%, and concentration difficulties in sixty-five point 6%. Vascular endothelial growth factor D and angiopoietin-1 remained elevated. Alpha-2-macroglobulin stayed high, and prothrombin time was prolonged in a subset of patients.

**Conclusions** Severe COVID-19 survivors show sustained angiogenic dysregulation and thromboinflammatory imbalance. Elevated vascular endothelial growth factor D and angiopoietin-1 indicate ongoing angiogenic perturbations, while increased alpha-2-macroglobulin and prolonged prothrombin time reflect persistent coagulation disturbances. These biomarkers may help identify long-term vascular alterations in post-COVID-19 syndrome.

**Keywords** Post-COVID-19 syndrome · Endothelial dysfunction · Hemostatic alterations · Angiogenic proteins · Long COVID · Biomarkers

---

Arlette Araceli Mora Mora Zetina, Erika Fabiola Ortega Ortega Sanchez and Brenda Saraí Zuñiga Zuñiga Ascencio contributed equally to this work.

Extended author information available on the last page of the article

## Introduction

The coronavirus disease 2019 (COVID-19) pandemic has resulted in a substantial global health burden, not only from acute infection but also from long-term sequelae affecting a subset of patients. Post-COVID-19 syndrome (PCS), also known as long COVID, is characterized by the persistence of symptoms such as fatigue, dyspnea, and cognitive disturbances for weeks or months after the resolution of acute SARS-CoV-2 infection [1]. The pathophysiology of PCS is complex and remains incompletely understood; however, accumulating evidence supports a central role for persistent thromboinflammatory processes, endothelial dysfunction, and dysregulated angiogenic signaling [2].

Endothelial dysfunction is a hallmark of acute COVID-19, driven by direct viral injury and an exaggerated inflammatory response [3]. It manifests as impaired vasodilation, increased vascular permeability, and a prothrombotic state, contributing to the high incidence of thromboembolic complications observed in severe disease [4]. In parallel, SARS-CoV-2 infection has been associated with alterations in angiogenic pathways, including imbalances in pro- and anti-angiogenic mediators, which may promote aberrant vascular remodeling and microvascular injury. While some studies suggest partial resolution of these abnormalities, others indicate that endothelial and angiogenic disturbances may persist for months or years, potentially contributing to the chronic manifestations of PCS [5].

Hemostatic abnormalities are also well documented in acute COVID-19, including elevated D-dimer levels, platelet hyperreactivity, and alterations in coagulation parameters such as prothrombin time (PT) and activated partial thromboplastin time (aPTT) [6]. These findings reflect a hypercoagulable and thromboinflammatory state closely linked to endothelial activation and immune dysregulation, and are associated with an increased risk of thrombotic events [7]. However, the long-term trajectory of these thromboinflammatory alterations in PCS remains poorly defined, with some studies reporting normalization and others demonstrating persistent abnormalities, particularly in patients with severe acute disease or underlying comorbidities [8].

Acute-phase proteins (APPs), including C-reactive protein (CRP), ferritin, haptoglobin, and alpha-2-macroglobulin, are key components of the systemic inflammatory response and have been associated with disease severity and outcomes in acute COVID-19 [9]. Beyond their role as inflammatory markers, several APPs regulate protease activity and coagulation pathways, thereby linking inflammation with thromboinflammation and vascular remodeling. The dynamics of these proteins in the post-acute phase

are less well characterized, but persistent alterations may reflect ongoing biological processes contributing to PCS pathogenesis [10]. Importantly, some of these mediators may also influence angiogenic signaling, further supporting their relevance in long-term vascular alterations.

The clinical manifestations of PCS are heterogeneous and affect multiple organ systems. Frequently reported symptoms include fatigue, dyspnea, chest pain, cognitive impairment (“brain fog”), and psychological disturbances such as anxiety and depression [11]. These symptoms can significantly impair quality of life and functional capacity. The underlying mechanisms are likely multifactorial, involving persistent immune activation, microvascular dysfunction, thromboinflammation, and dysregulated angiogenesis [12].

Despite the growing recognition of PCS as a distinct clinical entity, there is a paucity of data regarding the long-term evolution of angiogenic and thromboinflammatory biomarkers in affected patients [13]. Most studies have focused on the acute or subacute phases, with limited follow-up beyond six months. A better understanding of the persistence and interplay of these pathways is essential to elucidate the pathophysiology of PCS, identify patients at risk for chronic complications, and guide the development of targeted therapeutic strategies [14, 15].

The present study aims to address this knowledge gap by evaluating long-term changes in biomarkers reflecting angiogenic activity and thromboinflammatory processes in a well-characterized cohort of patients with PCS. Specifically, we compared biomarker levels during acute hospitalization and at long-term follow-up (3.5 years post-infection), and assessed their association with persistent clinical symptoms. Our objective is to provide insight into the vascular and inflammatory mechanisms underlying PCS and to identify candidate biomarkers associated with long-term outcomes in this population.

## Methodology

### Study design and setting

This research was conducted as an ambispective cohort study at the Instituto Nacional de Enfermedades Respiratorias (INER) in Mexico City, a national reference center for respiratory diseases. The primary objective was to evaluate the long-term evolution of endothelial dysfunction, hemostatic alterations, and acute-phase protein biomarkers in patients diagnosed with post-COVID-19 syndrome. The study included both retrospective and prospective components: clinical and laboratory data from the acute phase of COVID-19 hospitalization (2020) were retrieved from

institutional records, and new samples and clinical assessments were collected during the follow-up period between 2023 and 2024, corresponding to approximately 3.5 years post-infection. The study protocol was approved by the INER ethics committee, and all participants provided written informed consent prior to enrollment.

### Participants and eligibility criteria

The cohort comprised adult patients (aged  $\geq 18$  years) who had been hospitalized at INER in 2020 with a confirmed diagnosis of severe or critical COVID-19, as defined by national and World Health Organization criteria. Inclusion required a positive RT-PCR test for SARS-CoV-2 and availability for long-term follow-up. Exclusion criteria included pre-existing chronic inflammatory, autoimmune, or hematological disorders unrelated to COVID-19, as well as refusal or inability to provide informed consent. A total of 32 patients met these criteria and were included in the analysis. Demographic data, including age, sex, and relevant comorbidities, were extracted from the institutional database. The median age of participants was 50 years (interquartile range [IQR]: 43–60 years), with a sex distribution of 75% male and 25% female.

### Sample collection and biomarker measurement

For each participant, two peripheral blood samples were collected: one during acute hospitalization in 2020 (basal phase, BP) and another during the post-COVID follow-up visit in 2023–2024 (follow-up phase, FP). Blood samples were obtained in EDTA tubes and processed within two hours. Plasma was isolated by centrifugation and stored at  $-80^{\circ}\text{C}$  until analysis to preserve biomarker stability.

Endothelial dysfunction and angiogenic imbalance were assessed through the quantification of circulating angiogenic and endothelial activation markers, including soluble P-selectin (sP-selectin), vascular endothelial growth factor receptor 2 (VEGF-R2), vascular endothelial growth factor D (VEGF-D), and angiopoietin-1 (Ang-1). These biomarkers were selected based on their established role in endothelial activation, vascular permeability, and angiogenic signaling. Quantification was performed using a validated bead-based multiplex immunoassay (Invitrogen<sup>TM</sup> Coagulation 6-Plex Human ProcartaPlex<sup>TM</sup> Panel; catalog number 1553245), following the manufacturer's instructions.

To complement the evaluation of endothelial and angiogenic pathways, acute-phase proteins associated with systemic inflammation and vascular response, including alpha-2-macroglobulin and haptoglobin, were measured using a bead-based multiplex immunoassay (Bio-Plex

Pro<sup>TM</sup> Human Acute Phase Assay Panel, Bio-Rad; catalog number 171-A4010M).

Hemostatic parameters were assessed to characterize the interplay between endothelial dysfunction and coagulation. Prothrombin time (PT) and thrombin time (TT) was measured using an automated coagulation analyzer (ACL TOP 500 CTS<sup>TM</sup>, Werfen/Instrumentation Laboratory), and additional parameters such as D-dimer levels and platelet count were recorded when available.

All assays were performed in duplicate to ensure analytical reproducibility. Internal quality controls were included in each run, and laboratory personnel were blinded to clinical data and outcomes.

### Clinical data collection and symptom assessment

Clinical data from the acute phase were obtained from the INER COVID-19 database, including disease severity, length of hospital stay, and requirement for intensive care unit admission or mechanical ventilation.

During the follow-up phase, participants underwent a structured clinical evaluation conducted by trained physicians. A standardized questionnaire was used to assess persistent symptoms, categorized as cardiorespiratory (dyspnea, cough, odynophagia), neurocognitive (headache, concentration difficulties), and systemic (fatigue, myalgia, arthralgia). Functional status was evaluated using the modified Medical Research Council (mMRC) dyspnea scale.

All data were systematically recorded in electronic case report forms specifically designed for the study (Fig. 1).

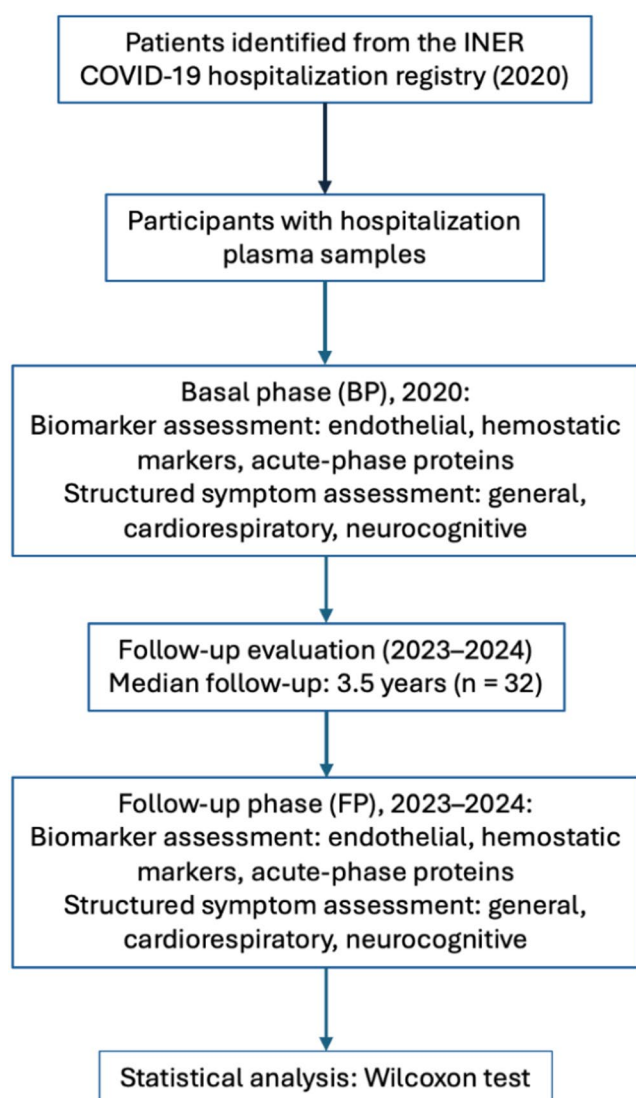
### Statistical analysis

Descriptive statistics were used to summarize demographic and clinical characteristics. Continuous variables were expressed as mean  $\pm$  standard deviation (SD) or median with interquartile range (IQR), depending on data distribution, while categorical variables were presented as frequencies and percentages.

Normality was assessed using the Shapiro–Wilk test. Paired comparisons of biomarker levels between BP and FP were performed using the Wilcoxon signed-rank test for non-normally distributed variables or paired t-test when appropriate.

Correlations between angiogenic biomarkers and persistent clinical symptoms were evaluated using Spearman's rank correlation coefficient. A two-sided  $p$ -value  $< 0.05$  was considered statistically significant.

All statistical analyses were conducted using SPSS software (version 25.0; IBM Corp., Armonk, NY, USA).



**Fig. 1** Study workflow: recruitment, sample collection during the acute phase (AP) and follow-up phase (FP), biomarker assessment, and statistical analysis

## Results

### Study population and clinical outcomes

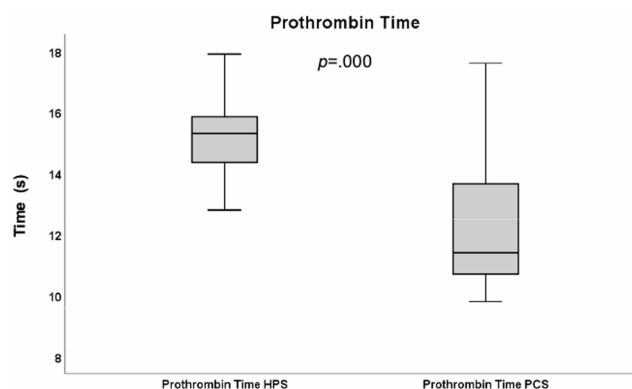
Our longitudinal cohort included 32 patients with severe or critical COVID-19 (median age 50 years, IQR: 43–60; 75% male) followed for a median of 3.5 years (2020–2024). The comorbidities observed were: 93.8% obesity, 34.4% type 2 diabetes, and 3.1% hypertension. Persistent post-acute symptoms were common: 65.6% of patients reported dyspnea, 62.5% experienced chronic fatigue, and 65.6% had cognitive impairment, particularly concentration difficulties, reflecting the multisystemic nature of post-COVID-19 syndrome (Table 1).

**Table 1** Baseline demographic characteristics, comorbidities, and symptom prevalence during acute phase and at 3.5-year follow-up in patients with post-COVID-19 syndrome ( $n=32$ )

| Characteristic             | Value              |                             |                  |
|----------------------------|--------------------|-----------------------------|------------------|
| Age, median (IQR), years   | 50 (43–60)         |                             |                  |
| Sex, n (%)                 |                    |                             |                  |
| Male                       | 24 (75.0)          |                             |                  |
| Female                     | 8 (25.0)           |                             |                  |
| Comorbidities, n (%)       |                    |                             |                  |
| Obesity                    | 30 (93.8)          |                             |                  |
| Type 2 diabetes            | 11 (34.4)          |                             |                  |
| Hypertension               | 1 (3.1)            |                             |                  |
| Symptom                    | Acute phase, n (%) | Follow-up (3.5 years) n (%) | <i>p</i> -value* |
| Cardiorespiratory          |                    |                             |                  |
| Dyspnea                    | 26 (81.3)          | 21 (65.6)                   | 0.245            |
| Cough                      | 24 (75.0)          | 8 (25.0)                    | <b>&lt;0.001</b> |
| Odynophagia                | 9 (28.1)           | 9 (28.1)                    | 1                |
| Neurocognitive             |                    |                             |                  |
| Headache                   | 13 (40.6)          | 11 (34.4)                   | 0.791            |
| Concentration difficulties | 0 (0.0)            | 21 (65.6)                   | <b>&lt;0.001</b> |
| General                    |                    |                             |                  |
| Fatigue                    | 7 (21.9)           | 20 (62.5)                   | <b>0.002</b>     |
| Myalgia                    | 16 (50.0)          | 15 (46.9)                   | 1                |
| Arthralgia                 | 16 (50.0)          | 12 (37.5)                   | 0.442            |

Fisher's exact test (two-tailed) comparing acute phase vs. follow-up. Statistically significant differences ( $p < 0.05$ ) are shown in bold

IQR, interquartile range



**Fig. 2** Hemostatic abnormalities: comparison of prothrombin time between the acute phase and the follow-up phase; Wilcoxon signed-rank test

### Hemostatic profile and coagulation dynamics

Non-parametric analysis showed that prothrombin time (PT) was significantly prolonged during the acute hospitalization phase (median of 15.3 s, IQR: 14.3–15.8), and although improved, remained elevated at follow-up with a median of 11.4 s (IQR: 10.6–13.8 Wilcoxon signed-rank test,  $p < 0.001$ ). This indicates persistent coagulation abnormalities beyond clinical recovery. (Fig. 2)

## Inflammatory and acute-phase proteins

Alpha-2-macroglobulin levels increased significantly from a median of 38.7 mg/dL (IQR: 0.91–69.6) during acute illness to 86.8 mg/dL (IQR: 50.2–138.1) at follow-up ( $p < 0.001$ ), indicating sustained, subclinical systemic inflammation. Haptoglobin remained elevated above baseline with median values 56.7 mg/dL (IQR: 22.6–315) of follow-up compared to 139.6 mg/dL (IQR: 31.4–315) during the acute phase ( $p = 0.02$ ). (Fig. 3)

## Endothelial dysfunction biomarkers

Markers of endothelial activation and vascular repair exhibited distinct longitudinal patterns. Soluble P-selectin (sP-Selectin) decreased significantly from a median of 121,700 pg/mL (IQR: 68,900–351,000) in the acute phase to 58,000 pg/mL (IQR: 23,000–141,000) at follow-up ( $p = 0.010$ ), indicating partial resolution of endothelial activation.

Conversely, VEGF-D showed a marked increase from a median of 10.4 pg/mL (IQR: 6.5–30.6) during acute illness to 75.2 pg/mL (IQR: 50.4–92.8) at follow-up ( $p < 0.001$ ), exceeding normal ranges and suggesting ongoing vascular injury, lymphangiogenesis and inflammation. VEGF receptor 2 (VEGF-R2) showed a slight decrease from 4,730 pg/mL (IQR: 3,200–7,550) to 4,240 pg/mL (IQR: 3,420–5,200)  $p = 0.045$ , which may indicate persistent endothelial activation, compensatory angiogenesis and residual chronic inflammation.

Angiopoietin-1 exhibited a non-significant upward trend, with median values rising from 465 pg/mL (IQR: 235–1,160) to 440 pg/mL (IQR: 324–586;  $p = 0.064$ ), possibly reflecting compensatory vascular remodeling. (Fig. 4)

## Clinical correlations

Spearman correlation analyses showed that VEGFR-2 levels were strongly and positively correlated with fibrinogen ( $r = 0.803$ ,  $p < 0.001$ ), thrombin time ( $r = 0.809$ ,  $p < 0.001$ ), and prothrombin time ( $r = 0.722$ ,  $p < 0.001$ ) (Fig. 5).

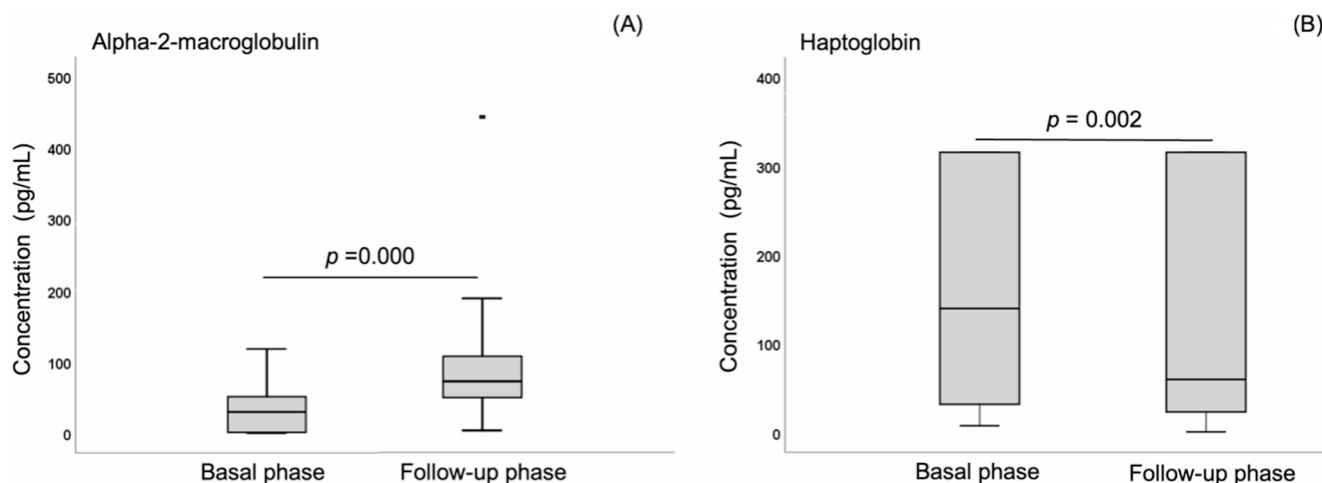
In a similar pattern, VEGF-D levels were also positively associated with fibrinogen ( $r = 0.815$ ,  $p < 0.001$ ), thrombin time ( $r = 0.838$ ,  $p < 0.001$ ), and prothrombin time ( $r = 0.759$ ,  $p < 0.001$ ), suggesting a consistent link between these angiogenic markers and coagulation parameters (Fig. 6).

Statistical analyses using Wilcoxon signed-rank tests confirmed that these longitudinal changes in PT, acute-phase proteins, and angiogenic markers were significant (all  $p < 0.01$ ). Importantly, VEGF-D concentrations correlated strongly with persistent dyspnea ( $r = 0.72$ ,  $p = 0.003$ ), while reductions in sP-Selectin were associated with improvements in fatigue scores ( $r = -0.58$ ,  $p = 0.018$ ), highlighting meaningful clinical-biomarker relationships.

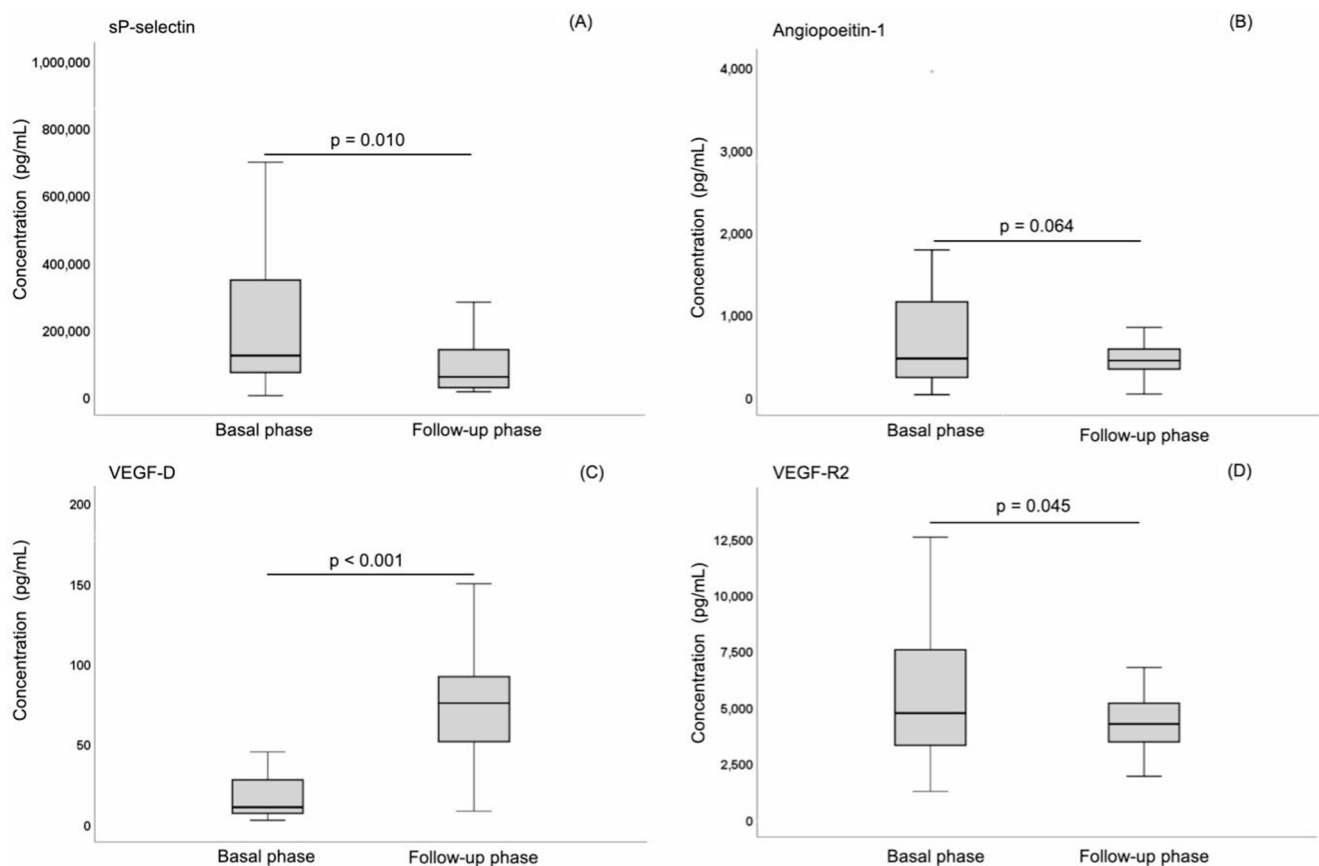
In summary, our findings reveal persistent coagulopathy characterized by prolonged PT, alongside chronic endotheliopathy evidenced by elevated VEGF-D and dynamic angiopoietin-1 levels.

## Discussion

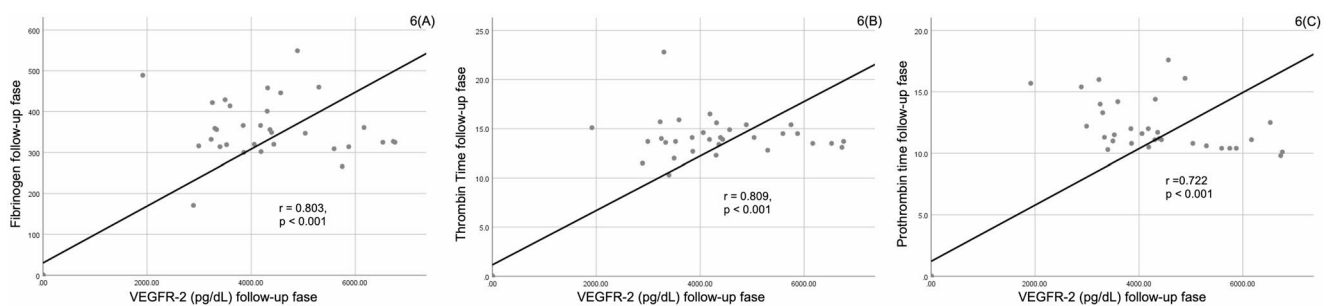
This study provides evidence of persistent elevated angiogenic proteins and hemostatic alterations, in a cohort of patients with PCS, 3.5 years after severe or critical COVID-19. The findings align with previous reports demonstrating that endothelial injury and microvascular dysfunction are central to PCS pathogenesis [16]. The sustained elevation of VEGF-D and angiopoietin-1 suggests ongoing endothelial activation or impaired repair mechanisms, which



**Fig. 3** Acute-phase proteins: comparison of  $\alpha_2$ -macroglobulin (A) and haptoglobin B between the acute phase and the follow-up phase (FU); Wilcoxon signed-rank test



**Fig. 4** Biomarkers of endothelial dysfunction: comparison of sP-selectin (A), angiopoietin-1 (B), VEGF-D (C), and VEGF-R2 (D) between the acute phase and the follow-up phase; Wilcoxon signed-rank test

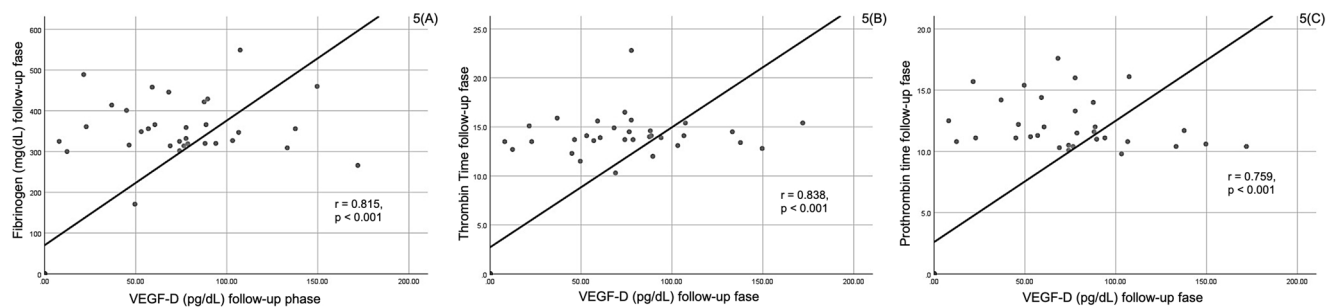


**Fig. 5** Figure 5. Spearman correlations between serum VEGFR-2 levels and hemostatic parameters during the follow-up phase. (A) Fibrinogen ( $r = 0.803$ ,  $p < 0.001$ ). (B) Thrombin time ( $r = 0.809$ ,  $p < 0.001$ ).

(C) Prothrombin time ( $r = 0.722$ ,  $p < 0.001$ ). Spearman's correlation coefficients ( $r$ ) and corresponding  $p$  values are shown in each panel.

may contribute to the chronic symptoms observed in these patients [17]. Specifically, we observed high rates of fatigue, dyspnea, and concentration difficulties, accompanied by sustained elevations in biomarkers of endothelial dysfunction (VEGF-D, angiopoietin-1), hemostatic alterations (prothrombin time), and acute-phase proteins (alpha-2-macroglobulin, haptoglobin) [18]. These findings underscore the chronic and multisystemic nature of post-COVID-19 syndrome and highlight the potential utility of these biomarkers as prognostic indicators for long-term complications.

Our results are consistent with and extend previous observations regarding the long-term sequelae of COVID-19. Multiple studies have documented persistent symptoms months after acute infection, with fatigue, dyspnea, and cognitive impairment being the most reported [19]. The high prevalence of these symptoms in our cohort, even after 3.5 years, suggests that post-COVID-19 syndrome may represent a chronic condition for a substantial subset of survivors, particularly those who experienced severe disease [20].



**Fig. 6** .Figure 6. Spearman correlations between serum VEGF-D levels and hemostatic parameters during the follow-up phase. (A) Fibrinogen ( $r = 0.815$ ,  $p < 0.001$ ). (B) Thrombin time ( $r = 0.838$ ,  $p < 0.001$ ).

(C) Prothrombin time ( $r = 0.759$ ,  $p < 0.001$ ). Spearman's correlation coefficients ( $r$ ) and corresponding  $p$  values are shown in each panel.

The sustained elevation of endothelial dysfunction markers such as VEGF-D and angiopoietin-1 is particularly noteworthy. Endothelial injury is recognized as a hallmark of acute COVID-19, contributing to microvascular thrombosis, inflammation, and multiorgan dysfunction [21]. While prior studies have shown that some endothelial markers normalize within months, our findings indicate that in a subset of patients, endothelial activation and injury may persist even 1 year after recovery [22]. This is in line with recent reports of ongoing microvascular pathology and impaired vascular repair in long COVID cohorts [23]. The chronic elevation of VEGF-D and angiopoietin-1 may reflect ongoing attempts at vascular remodeling or a failure to resolve endothelial damage, which could contribute to the persistence of symptoms such as fatigue and dyspnea [24].

In line with earlier studies demonstrating the activation of angiogenic pathways during and after SARS-CoV-2 infection, the current findings corroborate those reports. Specifically, Philippe et al. [25] demonstrated persistent angiogenic abnormalities in individuals with post-COVID conditions, supporting the concept that dysregulated vascular remodeling may contribute to long-term disease manifestations. In our cohort, sustained elevations of VEGF-D and angiopoietin-1 were observed up to 3.5 years after severe or critical COVID-19, substantially extending the duration of previously described angiogenic alterations. Together, these findings support the hypothesis that chronic endothelial activation and aberrant angiogenic signaling may represent long-lasting biological hallmarks of post-COVID-19 syndrome.

Hemostatic alterations, particularly prolonged prothrombin time, were also observed in our cohort at long-term follow-up. Coagulation abnormalities, including elevated D-dimer and PT, are well documented in the acute phase of COVID-19 and have been associated with increased risk of thromboembolic events [26]. Our data suggest that these alterations may not fully resolve in all patients, raising concerns about a sustained prothrombotic state and the potential for late thrombotic complications [27]. This finding

supports the need for ongoing risk assessment and possibly extended thromboprophylaxis in selected high-risk individuals, as suggested by recent guidelines [28].

The persistent elevation of acute-phase proteins, particularly alpha-2-macroglobulin and haptoglobin, further supports the hypothesis of chronic low-grade inflammation in post-COVID-19 syndrome. These proteins are well-established markers of systemic inflammation and have been linked to disease severity and prognosis in acute COVID-19 [29]. Their sustained elevation years after infection suggests that immune dysregulation and unresolved inflammatory processes may underlie the chronic symptoms observed in long COVID patients [30]. This is consistent with emerging evidence of persistent immune activation, autoimmunity, and dysregulated cytokine profiles in post-acute sequelae of COVID-19 [31].

## Originality and contribution to the field

This study is among the first to provide a comprehensive long-term evaluation (3.5 years) of endothelial, hemostatic, and inflammatory biomarkers in a well-characterized cohort of post-COVID-19 patients. While previous research has largely focused on the acute and subacute phases, our data demonstrate that biological alterations may persist far beyond the initial recovery period. The identification of VEGF-D [32], angiopoietin-1 [33], and alpha-2-macroglobulin [29] as potential prognostic markers for long-term complications is novel and may inform future risk stratification and management strategies. By linking persistent biomarker abnormalities with ongoing clinical symptoms, our findings also advance the understanding of the pathophysiological mechanisms underlying post-COVID-19 syndrome and highlight potential therapeutic targets for intervention [34].

## Limitations

Despite its strengths, this study has several limitations that should be acknowledged. First, the sample size is relatively small and derived from a single center, which may limit the generalizability of the findings. Second, the lack of a matched control group (e.g., patients with mild COVID-19 or uninfected individuals) precludes definitive conclusions about the specificity of the observed biomarker changes to post-COVID-19 syndrome. Third, the observational design does not allow for causal inferences regarding the relationship between biomarker alterations and clinical outcomes. Four, the reliance on self-reported symptoms may introduce recall bias, and the potential impact of comorbidities or concurrent medications was not fully explored.

Furthermore, VEGF-A, a key regulator of angiogenesis and vascular permeability, was not included in the biomarker panel; therefore, a comprehensive assessment of VEGF-related signaling pathways was not possible. Finally, while we assessed a panel of well-established biomarkers, other relevant pathways (e.g., autoantibodies [35], specific cytokines [36], or markers of autonomic dysfunction [37]) were not analyzed and warrant further investigation.

## Implications and future directions

The persistence of endothelial dysfunction, coagulation abnormalities, and inflammation in post-COVID-19 patients has important clinical and research implications. Clinically, our findings support the need for long-term follow-up and multidisciplinary management of individuals recovering from severe or critical COVID-19, with particular attention to cardiovascular, thrombotic, and neurocognitive complications [38]. Sustained elevations of VEGF-D [32], angiopoietin-1 [33], and alpha-2-macroglobulin highlight their potential utility as candidate biomarkers may facilitate early identification of patients at risk for chronic sequelae and guide personalized therapeutic strategies [39, 41].

From a clinical translation viewpoint, these results highlight the need for larger, multicenter studies with extended follow-up and appropriate control groups to validate and expand upon our findings [40]. Mechanistic studies exploring the drivers of persistent endothelial and immune activation, as well as interventional trials targeting these pathways, are warranted [31]. For instance, therapies aimed at restoring endothelial function, modulating inflammation, or addressing coagulation abnormalities may hold promise for alleviating the burden of post-COVID-19 syndrome [2]. Building on these observations, the emerging interest in VEGF-targeted therapies, as exemplified by the ongoing BASECOVID trial evaluating bevacizumab in post-COVID-19 syndrome [42],

underscores the need for a more comprehensive characterization of angiogenic pathways in long COVID. Future studies incorporating VEGF-A, VEGF-D33, VEGF receptor signaling, and other angiogenic mediators may further clarify the therapeutic relevance of angiogenic dysregulation in PCS. Furthermore, integrating biomarker profiling with advanced imaging and functional assessments could provide a more comprehensive understanding of the multi-systemic impact of long COVID.

In summary, our study contributes to the growing body of evidence that post-COVID-19 syndrome is a complex, chronic condition with significant clinical and biological sequelae. The persistence of endothelial, hemostatic, and inflammatory abnormalities years after acute infection underscores the need for ongoing vigilance and innovation in both clinical care and research. By characterizing the long-term dynamics of key angiogenic and thromboinflammatory biomarkers, this study provides further evidence that persistent vascular dysfunction may contribute to the long-term consequences of severe COVID-19. These findings underscore the need for larger prospective studies to elucidate the mechanisms driving sustained vascular, inflammatory, and coagulation-related alterations following infection. Ultimately, a deeper understanding of these pathways may enhance risk assessment and support the identification of novel therapeutic targets for patients with PCS.

## Conclusions

Our study demonstrates that a substantial proportion of patients with post-COVID-19 syndrome exhibit persistent angiogenic and thromboinflammatory alterations more than three years after severe or critical COVID-19. Sustained elevations of VEGF-D and angiopoietin-1 suggest ongoing dysregulation of angiogenic pathways, whereas increased alpha-2-macroglobulin levels and prolonged prothrombin time indicate persistent disturbances in inflammatory and coagulation-related processes. These biological abnormalities correlate with a substantial burden of persistent symptoms, supporting the hypothesis that long-term vascular dysfunction may contribute to the pathophysiology of post-COVID-19 syndrome. Additionally, VEGF-D, angiopoietin-1, and alpha-2-macroglobulin could serve as valuable biomarkers for these changes. Further research is needed to clarify these findings and explore their potential clinical implications.

**Acknowledgements** Thanks to the Red MEDICI social service program from FESI, UNAM, for their invaluable support in allowing the students to complete their social service at Instituto Nacional Enfermedades Respiratorias.

**Author contributions** Conceptualization: M.C.L., M.E.H.; Methodology: M.C.L., A.A.M., E.F.O., V.H.A., L.M.T., A.A.G., B.S.Z., J.E.M., A.M.M.; Supervision, A.R.C., M.C.L., M.E.H., A.B.S., B.S.Z., P.L.S., J.L.A., J.G.M., L.M.T.; Design of figures and tables, A.A.M., E.F.O., M.O.G., M.R.F., D.M.B., A.H.I.; Writing-Original draft preparation and Writing-Review and editing: A.A.G., R.H.Z., J.Ch.A., A.R.C., E.B.V., D.J.C. All authors reviewed the manuscript.

**Funding** This work was carried out by medical specialists from different Institutes dependent on the government of Mexico, we did not receive additional funds for the current research than our assigned salary.

**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare no competing interests.

**Ethical approval** The Institutional Review Board approved the study (Approval E-17-20). Given the prospective nature of the study, we obtained informed consent of each patient.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

## References

- Soriano JB, Murthy S, Marshall JC, Relan P, Diaz JV (2022) A clinical case definition of post-COVID-19 condition by a Delphi consensus. *Lancet Infect Dis* 22(4):e102–e107. [https://doi.org/10.1016/S1473-3099\(21\)00703-9](https://doi.org/10.1016/S1473-3099(21)00703-9)
- Pretorius E, Vlok M, Venter C et al (2021) Persistent clotting protein pathology in long COVID/post-acute sequelae of COVID-19 (PASC) is accompanied by increased levels of antiplasmin. *Cardiovasc Diabetol* 20(1):172. <https://doi.org/10.1186/s12933-021-01359-7>
- Varga Z, Flammer AJ, Steiger P et al (2020) Endothelial cell infection and endotheliitis in COVID-19. *Lancet* 395(10234):1417–1418. [https://doi.org/10.1016/S0140-6736\(20\)30937-5](https://doi.org/10.1016/S0140-6736(20)30937-5)
- Ackermann M, Verleden SE, Kuehnel M et al (2020) Pulmonary vascular endothelialitis, thrombosis, and angiogenesis in Covid-19. *N Engl J Med* 383(2):120–128. <https://doi.org/10.1056/NEJMoa2015432>
- Charfeddine S, Ibn Hadj Amor H, Jdidi J et al (2021) Long COVID-19 syndrome: is it related to microcirculation and endothelial dysfunction? Insights from TUN-EndCOV study. *Front Cardiovasc Med* 8:745758. <https://doi.org/10.3389/fcvm.2021.745758>
- Tang N, Li D, Wang X, Sun Z (2020) Abnormal coagulation parameters are associated with poor prognosis in patients with novel coronavirus pneumonia. *J Thromb Haemost* 18(4):844–847. <https://doi.org/10.1111/jth.14768>
- Connors JM, Levy JH (2020) COVID-19 and its implications for thrombosis and anticoagulation. *Blood* 135(23):2033–2040. <https://doi.org/10.1182/blood.2020006000>
- Fogarty H, Townsend L, Morrin H et al (2021) Persistent endotheliopathy in the pathogenesis of long COVID syndrome. *J Thromb Haemost* 19(10):2546–2553. <https://doi.org/10.1111/jth.15490>
- Chen G, Wu D, Guo W et al (2020) Clinical and immunological features of severe and moderate coronavirus disease 2019. *J Clin Invest* 130(5):2620–2629. <https://doi.org/10.1172/JCI137244>
- Phetsouphanh C, Darley DR, Wilson DB et al (2022) Immunological dysfunction persists for 8 months following initial mild-to-moderate SARS-CoV-2 infection. *Nat Immunol* 23(2):210–216. <https://doi.org/10.1038/s41590-021-01113-x>
- Huang I, Pranata R, Lim MA, Oehadian A, Alisjahbana B (2020) C-reactive protein, procalcitonin, D-dimer, and ferritin in severe coronavirus disease 2019: a meta-analysis. *Ther Adv Respir Dis* 14:1753466620937175. <https://doi.org/10.1177/1753466620937175>
- Davis HE, Assaf GS, McCorkell L et al (2021) Characterizing long COVID in an international cohort: 7 months of symptoms and their impact. *EClinicalMedicine* 38:101019. <https://doi.org/10.1016/j.eclinm.2021.101019>
- Proal AD, VanElzakker MB (2021) Long COVID or post-acute sequelae of COVID-19 (PASC): an overview of biological factors that may contribute to persistent symptoms. *Front Microbiol* 12:698169. <https://doi.org/10.3389/fmicb.2021.698169>
- Townsend L, Fogarty H, Dyer A et al (2021) Prolonged elevation of D-dimer levels in convalescent COVID-19 patients is independent of the acute phase response. *J Thromb Haemost* 19(4):1064–1070. <https://doi.org/10.1111/jth.15267>
- Raveendran AV, Jayadevan R, Sashidharan S (2021) Long COVID: an overview. *Diabetes Metab Syndr* 15(3):869–875. <https://doi.org/10.1016/j.dsx.2021.04.007>
- Goshua G, Pine AB, Meizlish ML et al (2020) Endotheliopathy in COVID-19-associated coagulopathy: evidence from a single-centre, cross-sectional study. *Lancet Haematol* 7(8):e575–e582. [https://doi.org/10.1016/S2352-3026\(20\)30216-7](https://doi.org/10.1016/S2352-3026(20)30216-7)
- Patel MA, Knauer MJ, Nicholson M et al (2022) Elevated vascular transformation blood biomarkers in long COVID indicate angiogenesis as a key pathophysiological mechanism. *Mol Med* 28(1):122. <https://doi.org/10.1186/s10020-022-00548-8>
- Kruger A, Vlok M, Turner S et al (2022) Proteomics of fibrin amyloid microclots in long COVID/post-acute sequelae of COVID-19 (PASC) shows many entrapped pro-inflammatory molecules that may also contribute to a failed fibrinolytic system. *Cardiovasc Diabetol* 21(1):190. <https://doi.org/10.1186/s12933-022-01623-4>
- Taquet M, Dercon Q, Luciano S, Geddes JR, Husain M, Harrison PJ (2021) Incidence, co-occurrence, and evolution of long-COVID features: a 6-month retrospective cohort study of 273,618 survivors of COVID-19. *PLoS Med* 18(9):e1003773. <https://doi.org/10.1371/journal.pmed.1003773>
- Sudre CH, Murray B, Varsavsky T et al (2021) Attributes and predictors of long COVID. *Nat Med* 27(4):626–631. <https://doi.org/10.1038/s41591-021-01292-y>
- Evans PC, Rainger GE, Mason JC et al (2020) Endothelial dysfunction in COVID-19: a position paper of the ESC working group for atherosclerosis and vascular biology, and the ESC council of basic cardiovascular science. *Cardiovasc Res* 116(14):2177–2184. <https://doi.org/10.1093/cvr/cvaa230>

22. Fan BE, Wong SW, Sum CLL et al (2022) Hypercoagulability, endotheliopathy, and inflammation approximating 1 year after recovery: assessing the long-term outcomes in COVID-19 patients. *Am J Hematol* 97(7):915–923. <https://doi.org/10.1002/ajh.26575>
23. Osiaevi I, Schulze A, Evers G et al (2023) Persistent capillary rarefaction in long COVID syndrome. *Angiogenesis* 26(1):53–61. <https://doi.org/10.1007/s10456-022-09850-9>
24. Fogarty H, Ward SE, Townsend L et al (2022) Sustained VWF-ADAMTS13 axis imbalance and endotheliopathy in long COVID syndrome is related to immune dysfunction. *J Thromb Haemost* 20(10):2429–2438. <https://doi.org/10.1111/jth.15830>
25. Philippe A, Günther S, Rancic J, Cavagna P, Renaud B, Gendron N, Mousseaux E, Hua-Huy T, Reverdito G, Planquette B, Sanchez O, Gaussem P, Salmon D, Diehl JL, Smadja DM (2024) VEGF-A plasma levels are associated with impaired DLCO and radiological sequelae in long COVID patients. *Angiogenesis* 27(1):51–66. <https://doi.org/10.1007/s10456-023-09890-9>
26. Al-Samkari H, Karp Leaf RS, Dzik WH et al (2020) COVID-19 and coagulation: bleeding and thrombotic manifestations of SARS-CoV-2 infection. *Blood* 136(4):489–500. <https://doi.org/10.1182/blood.2020006520>
27. Korompoki E, Gavriatopoulou M, Fotiou D, Ntanasis-Stathopoulos I, Dimopoulos MA, Terpos E (2022) Late-onset hematological complications post COVID-19: an emerging medical problem for the hematologist. *Am J Hematol* 97(1):119–128. <https://doi.org/10.1002/ajh.26384>
28. Schulman S, Arnold DM, Bradbury CA et al (2024) 2023 ISTH update of the 2022 ISTH guidelines for antithrombotic treatment in COVID-19. *J Thromb Haemost* 22(6):1779–1797. <https://doi.org/10.1016/j.jth.2024.02.011>
29. Akbasheva OE, Mitrofanova DK, Spirina LV et al (2023) Alpha-2 macroglobulin activity in SARS-CoV-2-induced infection and in the post-COVID-19 period. *Biomed Khim* 69(4):240–247. <https://doi.org/10.18097/PBMC20236904240>
30. Phetsouphanh C, Darley D, Wilson DB et al (2021) Immunological dysfunction persists for 8 months following initial mild-to-moderate SARS-CoV-2 infection. *medRxiv* [Preprint]. 3. <https://doi.org/10.1101/2021.06.01.21257759>
31. Klein J, Wood J, Jaycox JR et al (2023) Distinguishing features of long COVID identified through immune profiling. *Nature* 623(7985):139–148. <https://doi.org/10.1038/s41586-023-06651-y>
32. Kong Y, Han J, Wu X, Zeng H, Liu J, Zhang H (2020) VEGF-D: a novel biomarker for detection of COVID-19 progression. *Crit Care* 24(1):373. <https://doi.org/10.1186/s13054-020-03079-y>
33. Yamga E, Soulé A, Piché A, Emad A, Durand M, Rousseau S (2023) Validation of ANG-1 and P-SEL as biomarkers of post-COVID-19 conditions using data from the Biobanque québécoise de la COVID-19 (BQC-19). *Clin Proteom* 20(1):63. <https://doi.org/10.1186/s12014-023-09436-7>
34. Su Y, Yuan D, Chen DG et al (2022) Multiple early factors anticipate post-acute COVID-19 sequelae. *Cell* 185(5):881–895e20. <https://doi.org/10.1016/j.cell.2022.01.014>
35. Wilhelm F, Cadamuro J, Mink S (2026) Autoantibodies in long COVID: a systematic review. *Lancet Infect Dis* 26(4):e220–e230. [https://doi.org/10.1016/S1473-3099\(25\)00411-6](https://doi.org/10.1016/S1473-3099(25)00411-6)
36. Jernbom AF, Skoglund L, Pin E et al (2024) Prevalent and persistent new-onset autoantibodies in mild to severe COVID-19. *Nat Commun* 15(1):9288. <https://doi.org/10.1038/s41467-024-53356-5>
37. Dani M, Dirksen A, Taraborrelli P et al (2021) Autonomic dysfunction in long COVID: rationale, physiology and management strategies. *Clin Med (Lond)* 21(1):e63–e67. <https://doi.org/10.7861/clinmed.2020-0896>
38. Nalbandian A, Sehgal K, Gupta A et al (2021) Post-acute COVID-19 syndrome. *Nat Med* 27(4):601–615. <https://doi.org/10.1038/s41591-021-01283-z>
39. Davis HE, McCorkell L, Vogel JM, Topol EJ, Long COVID (2023) major findings, mechanisms and recommendations. *Nat Rev Microbiol* 21(3):133–146. <https://doi.org/10.1038/s41579-022-00846-2>
40. Michelen M, Manoharan L, Elkheir N et al (2021) Characterising long COVID: a living systematic review. *BMJ Glob Health* 6(9):e005427. <https://doi.org/10.1136/bmjgh-2021-005427>
41. Raman B, Cassar MP, Tunnicliffe EM et al (2021) Medium-term effects of SARS-CoV-2 infection on multiple vital organs, exercise capacity, cognition, quality of life and mental health, post-hospital discharge. *EClinicalMedicine* 31:100683. <https://doi.org/10.1016/j.eclinm.2020.100683>
42. National Library of Medicine (US) BASECOVID: Bevacizumab in long COVID patients with severe fatigue (BASECOVID) [Internet]. Bethesda (MD): National Library of Medicine (US); 2025 [cited 2026 Jul 5]. <https://clinicaltrials.gov/study/NCT07089719>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

## Authors and Affiliations

Arlette Araceli Mora Zetina<sup>1,11</sup> · Erika Fabiola Ortega Sanchez<sup>1,11</sup> · Brenda Saraí Zuñiga Ascencio<sup>2,12</sup> · Armando Roberto Castorena Maldonado<sup>12</sup> · Víctor Hugo Ahumada Topete<sup>3,12</sup> · Luz María Torres Espíndola<sup>4</sup> · Arnoldo Aquino Gálvez<sup>5</sup> · Rosario Fernández Plata<sup>1,8</sup> · Anjarath Lorena Higuera Iglesias<sup>1,8</sup> · David Martínez Briseño<sup>1,8</sup> · Aranza Bernal Silva<sup>1</sup> · Misael Osmar García Martín<sup>12,13</sup> · José Eduardo Márquez García<sup>6</sup> · Angelica Moncada Morales<sup>6</sup> · Jaime Chávez Alderete<sup>7</sup> · Rafael Hernandez Zenteno<sup>8,12</sup> · Eduardo Becerril Vargas<sup>9,12</sup> · Daniela Josefina Cataneo Piña<sup>10,12</sup> · José Leopoldo Aguilar Faisal<sup>11,14</sup> · Jazmín García Machorro<sup>11,14</sup> · Pedro López Sánchez<sup>11,14</sup> · María Elena Hernandez Campos<sup>11,14</sup> · Manuel Castillejos López<sup>1,8,13</sup> 

✉ María Elena Hernandez Campos  
mayehc@hotmail.com

✉ Manuel Castillejos López  
mcastillejos@gmail.com

- <sup>1</sup> Laboratorio de Investigación en Epidemiología e Infectología, Departamento de Investigación en EPOC y Tabaquismo, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México
- <sup>2</sup> Laboratorio Clínico, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México
- <sup>3</sup> Clínica de Inmunocompromiso por Enfermedades Infecciosas del Centro de Investigación por Enfermedades Infecciosas, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México
- <sup>4</sup> Laboratorio de Farmacología, Instituto Nacional de Pediatría, Av. Insurgentes Sur 3700-Letra C, 04530 Ciudad de México, México
- <sup>5</sup> Departamento de Investigación en Fibrosis, Laboratorio de Biología Molecular, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México
- <sup>6</sup> Laboratorio de Secuenciación Viroológica y Biología Molecular, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>7</sup> Departamento de investigación en Hiperreactividad Bronquial, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>8</sup> Centro de Investigación en tabaquismo y EPOC, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>9</sup> Laboratorio de Microbiología Clínica, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>10</sup> Coordinación de Geriatría y Cuidados Paliativos, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>11</sup> Programa de Maestría en Ciencias de la Salud, Instituto Politécnico Nacional, Salvador Díaz Mirón esq. Plan de San Luis S/N, Casco de Santo Tomas, 11340 Ciudad de México, México

<sup>12</sup> Dirección médica del Instituto Nacional de Enfermedades Respiratorias, Ismael Cosío Villegas, México, Calzada de Tlalpan, 14080 Ciudad de México 4502, México

<sup>13</sup> Coordinación de Infectología. Unidad de Epidemiología Hospitalaria e Infectología, Instituto Nacional de Enfermedades Respiratorias, “Ismael Cosío Villegas”, Calzada de Tlalpan 4502, 14080 Ciudad de México, México

<sup>14</sup> Instituto Politécnico Nacional, Escuela Superior de Medicina, Mexico city, Mexico